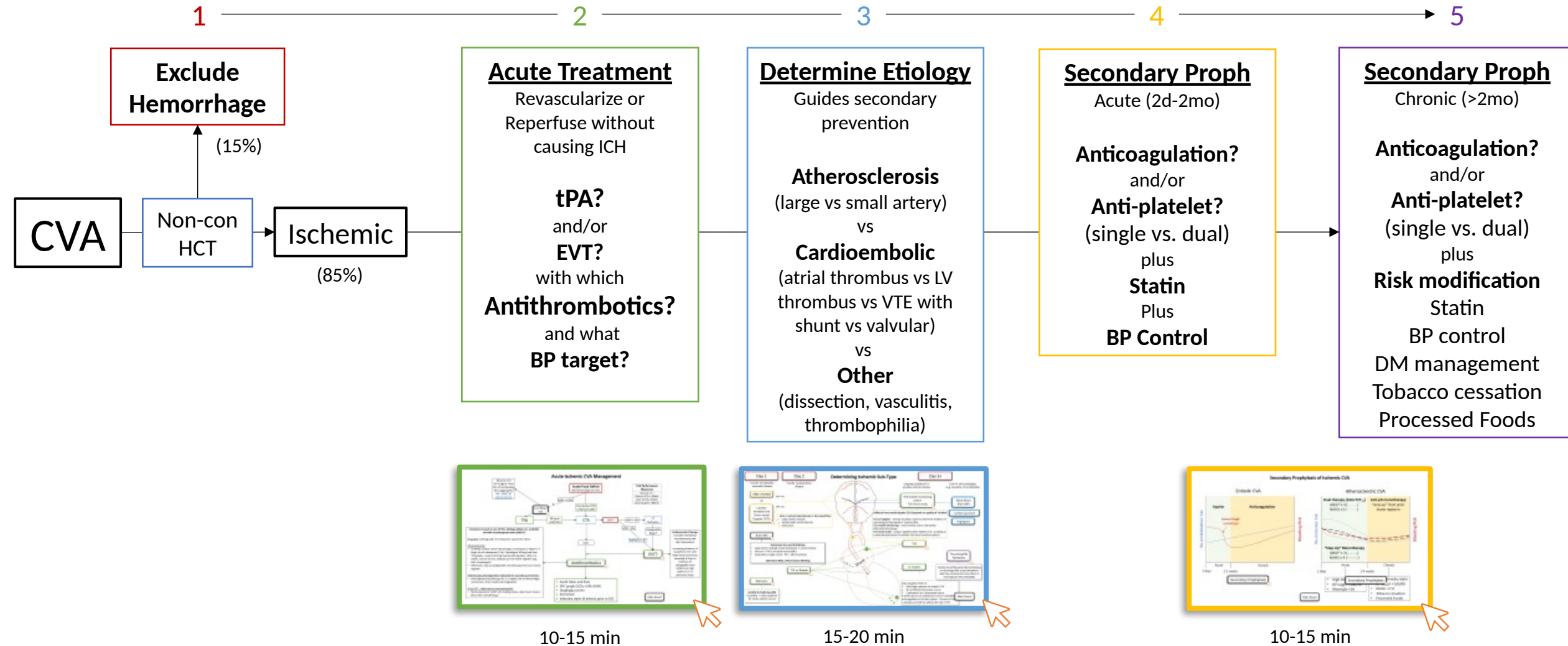


# 5-step Approach to Evaluating and Managing Ischemic CVAs



# Acute Ischemic CVA Management

NO prior ICH  
NO surgery <3mo  
NO active bleeding  
NO coagulopathy  
BP < 185/110  
[Link to Full List](#)

**Acute Focal Deficit**  
(No hemorrhage on HCT)

## CVA Performance Measures

|                       |
|-----------------------|
| Door to CT <          |
| Door to CTA < 40min   |
| Door to tPA < 30min   |
| Door to groin < 90min |

LKW < 4.5hr

Permissive HTN  
<220/120 x48hr

Low bleed risk

TPA

BP goal  
<185/110

CTA

LVO+

LKW > 6hr

CT Perfusion

LKW < 6hr

Salvageable Brain?

ASPECTS ≥ 6?

EVT?

LVO-

Antithrombotics

## Endovascular Therapy

Includes mechanical thrombectomy and stent placement.

Increasing evidence of benefits to EVT with large vessel occlusions, especially if there is evidence of salvageable brain either by a high ASPECTs or CT perfusion study.

Decision is based on use of tPA, etiology (athero vs. embolic) and risk of subsequent event (ABCD<sup>2</sup>)

If s/p tPA: nothing x24h, if no bleed on repeat HCT, then:

### Atherosclerotic:

- If, NIHSS ≤ 3 (low risk of hemorrhagic conversion) or ABCD<sup>2</sup> ≥ 4 (high risk of subsequent CVA): Clopidogrel 300mg load then 75mg daily + Aspirin 325mg load and 81mg daily. After 2-4 weeks, narrow to one, ramped up from home regimen (e.g. ASA->clopidogrel)
- Otherwise, step up antiplatelet monotherapy from prior home regimen

### Embolic plus anticoagulation indicated for secondary prevention

- Anti-platelet monotherapy for 1-2 weeks (risk of hemorrhagic conversion), then initiate anticoagulation

### If s/p EVT - deferring to interventionalist

- Stent placement: DAPT with loading doses, daily lower doses ≥ 6mo, then monotherapy

- Avoid tubes and lines
- DVT proph (SCDs <24h if tPA)
- Dysphagia screen
- PT/OT/SLP
- Intensive statin (if athero) prior to D/C

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## Standard Diagnostics

Eval for immediately reversible disease

Eval for cardioembolic disease

H&N CTA/MRA

GFR >45

OR

Carotid, Vertebral and Trans-cranial Doppler (TCD)

GFR <45

**Intra or extracranial stenosis or decreased flow**

- Large vessel occlusion
- Symptomatic carotid stenosis
- Dissections

Brain MRI

### 1. Size and Distribution suggests etiology

- Large and/or multiple vessel distribution ≈ cardio-embolic
- Lacunar ≈ HTN or amyloid vasculopathy
- Small and/or single vessel + RFs ≈ atherosclerosis

### 2. Determine Risks of Post-infarct Bleeding

Telemetry

**Identify Arrhythmia (afib)**

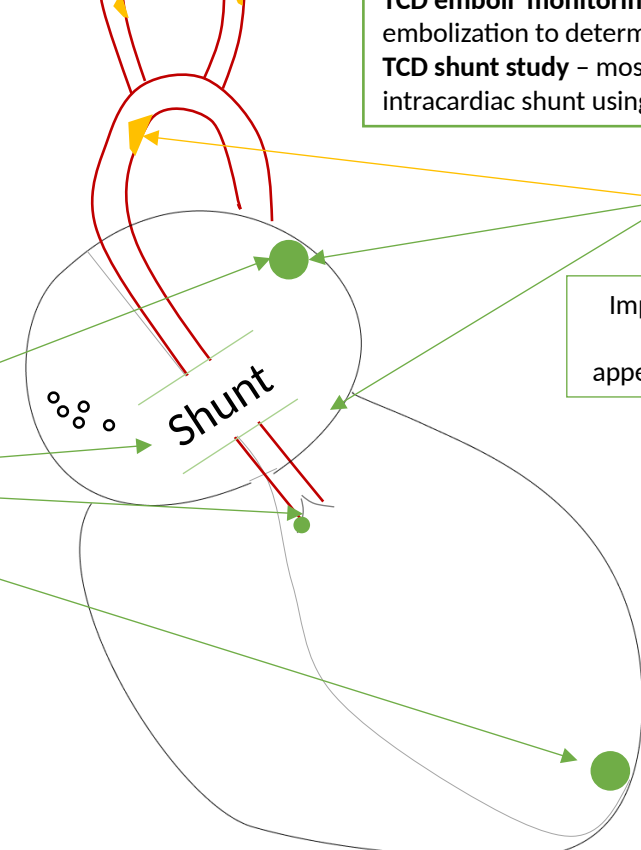
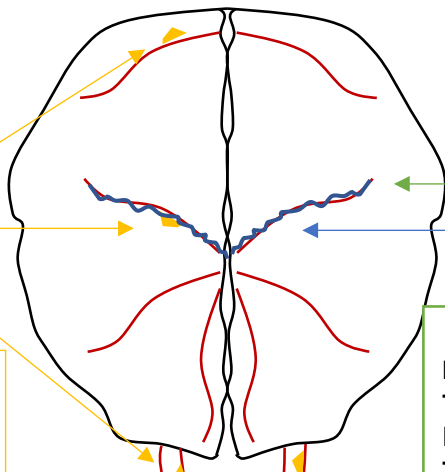
If positive -> raises suspicion for cardio-embolic source

TTE w/ Bubble

**All patients with new concerns or no recent study**

- Intracardiac thrombus
  - Atrial appendage in atrial fib
  - LV thrombus with apical akinesis
  - Mechanical MV or AV thrombus
- MV or AV rheumatic or infective endocarditis
- Right to Left shunt

## Determining Ischemic Sub-Type



## Specialized Diagnostics

Ongoing evaluation for possible embolic disease

Eval for other etiologies (e.g. vasculitis, thrombophilia)

TCD emboli monitoring and/or TCD shunt study

Black Blood Brain MRI

Lumbar puncture

Angiogram

**Utility depends on 'window' quality**

**Each modality answers different clinical questions:**

**TCD w/ duplex** - directly visualizes vessel and flow to identify local obstruction and inform severity of adjacent disease

**TCD emboli monitoring** - Localizes ongoing artery-artery embolization to determine urgency for intervention.

**TCD shunt study** - most sensitive test for identifying intracardiac shunt using IV infusion of agitated saline

TEE

Improved visualization of the mitral valve, left atrial appendage and ascending aorta

Thrombophilia Evaluation

Testing should be guided by hematology or neurology after acute thrombotic state has resolved and only if there is mod-high pre-test probability

LE Duplex

**Only indicated if:**

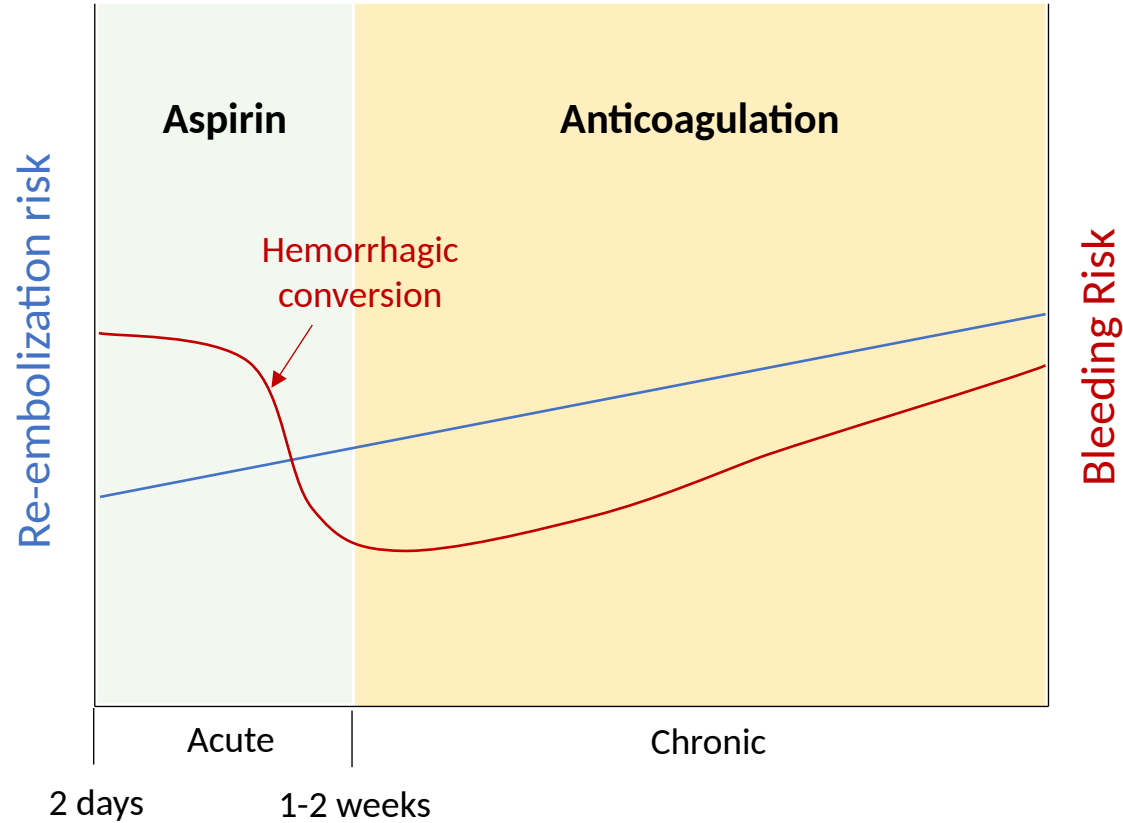
1. Mod-high suspicion of cardioembolic etiology
2. No identified intracardiac source
3. Confirmed intracardiac shunt

**If all the above**, anticoagulation preferred over closure or IVC filter to prevent VTE and CVA

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# Secondary Prophylaxis of Ischemic CVA

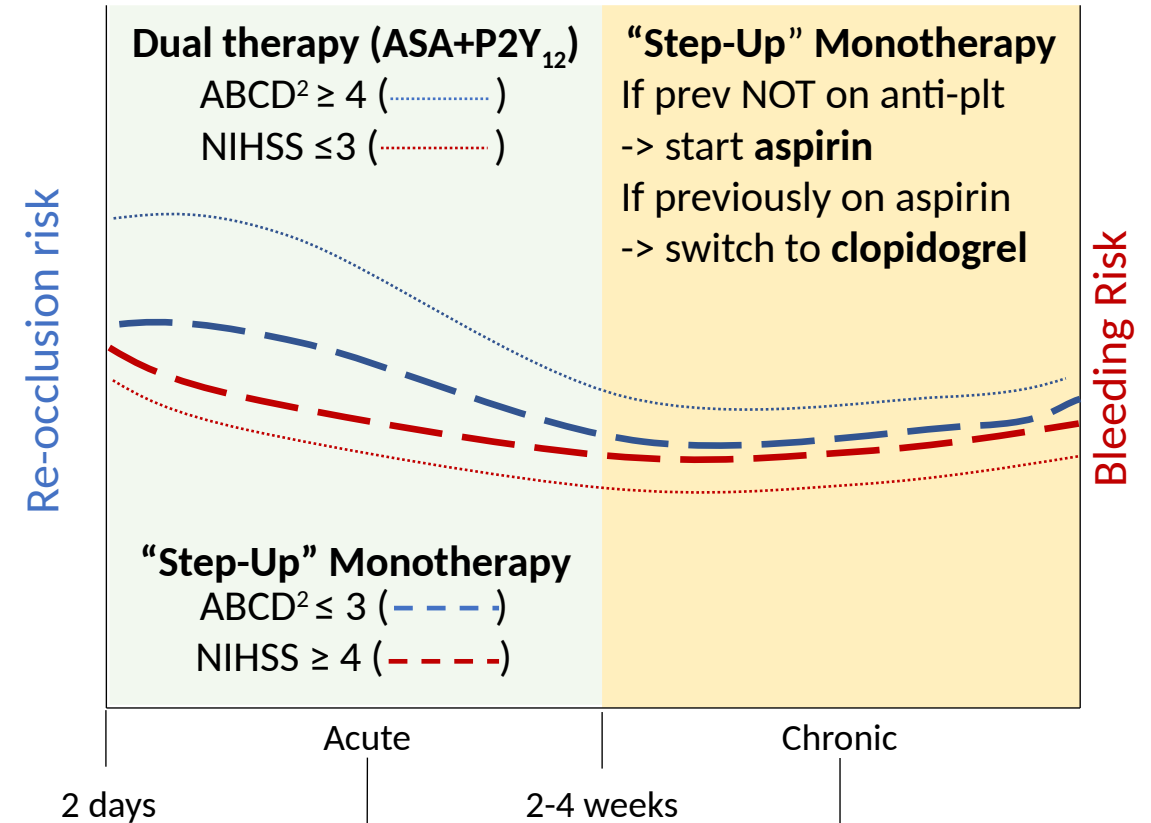
## Embolic CVA



Secondary Prophylaxis

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## Atherosclerotic CVA



- Secondary Prophylaxis
- High intensity statin
  - BP target < 140/90
  - Blood glu < 180
  - HbA1c < 7.0
  - Tobacco cessation
  - Processed Foods

# 2021–2024 Guideline Update

Standing 5-step framework is current; AHA/ASA refinements since 2020:

IV thrombolysis: tenecteplase (TNK) 0.25 mg/kg is now a Class 1 alternative to alteplase 0.9 mg/kg — non-inferior, simpler single-bolus, possibly superior for LVO (2024 AHA/ASA acute ischemic stroke guideline).

Thrombectomy expanded to LARGE-CORE infarcts: anterior LVO, 6–24 h, NIHSS  $\geq 6$ , ASPECTS 3–5, age  $< 80$  — Class 1 (SELECT2, ANGEL-ASPECT, NEJM 2023). EVT to 24 h with mismatch already established (DAWN/DEFUSE-3); basilar-occlusion EVT now supported (ATTENTION/BAOCHE 2022).

Secondary prevention (2021 AHA/ASA, Kleindorfer): DAPT clopidogrel+aspirin for 21 days (then mono) for minor stroke NIHSS  $\leq 3$  or high-risk TIA ABCD2  $\geq 4$  (CHANCE/POINT); ticagrelor+aspirin  $\times 30$  d an alternative (THALES). LDL-C goal  $< 70$  mg/dL with high-intensity statin  $\pm$  ezetimibe/PCSK9.

PFO closure is reasonable for cryptogenic embolic stroke in selected patients  $< 60$  y (RESPECT/CLOSE/REDUCE).

# Take-Home Points

Work every ischemic stroke through 5 steps: exclude hemorrhage (noncontrast HCT) → acute reperfusion (tPA/TNK ± EVT without causing ICH) → determine etiology → secondary prophylaxis → risk-factor modification.

Reperfusion is time-critical: IV thrombolysis if LKW <4.5 h and no contraindication (TNK 0.25 mg/kg or alteplase); thrombectomy for anterior LVO with salvageable brain, now including large-core in extended windows.

Etiology drives prevention: atherosclerotic → antiplatelet (DAPT 21 d for minor stroke/high-risk TIA, then mono); cardioembolic → anticoagulation after the hemorrhagic-conversion window.

Acute BP is permissive (<220/120 ×48 h; <185/110 if thrombolysed); long-term target <130/80. LDL <70, A1c <7, tobacco cessation, glucose <180 acutely.

# Attribution & Changelog

Adapted from TeachIM 'Ischemic CVA (Cerebrovascular Accident)' (Brandon Fainstad MD; expert review pending), pub. 2020.  
[teachim.org/teaching\\_material/ischemic-cva-cerebrovascular-accident/](https://teachim.org/teaching_material/ischemic-cva-cerebrovascular-accident/)

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Surgical corrections: 'emobli'→'emboli' (slide 3 TCD); 'asprin'→'aspirin' (slide 4); acute glucose target 'Blood glu <18'→'<180' mg/dL (slide 4 — inferred missing zero, flagged for human review).

Added (source-cited): 2024 AHA/ASA acute stroke guideline (TNK Class 1, large-core EVT SELECT2/ANGEL-ASPECT); 2021 AHA/ASA secondary-prevention guideline (DAPT 21 d, ticagrelor alt, LDL <70, PFO closure).

Appended: this update slide, take-home slide, attribution slide. All original animated/click-to-reveal builds preserved.

Optimization date 2026-06-10. Educational use only; clinician-review before teaching.